Lecture 8

Instrumental Variables & DML-IV

ECON6083: Machine Learning in Economics

Today's Roadmap

#	TOPIC	FOCUS
1	Endogeneity Reminder	Why IV, when to use
2	Classical IV & LATE	Compliers, intuition, limitations
3	High-Dimensional Challenge	When classical IV fails
4	JIVE and Regularized IV	Jackknife methods, sparse vs dense
5	DML-IV Theory	Orthogonality, cross-fitting for controls
6	GRF-IV for Heterogeneity	Who benefits from treatment
7	Applications	Institutions, 401(k), AI demand

Part 1

The Endogeneity Problem

What is Endogeneity?

Structural equation: $Y = \theta_0 D + X'\beta + \epsilon$

Endogeneity: $Cov(D, \epsilon) \neq 0$ — treatment correlated with unobservables

Three sources:

SOURCE	EXAMPLE	DIRECTION
Omitted variables	Ability affects education and earnings	Usually upward
Simultaneity	Price and quantity jointly determined	Ambiguous
Measurement error	Misreported schooling attenuates effect	Downward

> ***Consequence***: *OLS $\hat{\theta}_{OLS}$ is **inconsistent** — need instruments!*

The IV Core Idea

Instrument Z: A variable that satisfies three conditions:

1. **Relevance**: Correlated with treatment ($Z \to D$)
2. **Exclusion**: Affects outcome only through treatment ($Z \nrightarrow Y$ except via D)
3. **Independence**: As-good-as-randomly assigned ($Z \perp \epsilon$)

DAG intuition:

```
    Z → D → Y
        ↑   ↑
        └───┘
  (confounding blocked)
```

> *"**Intuition**: Use only the variation in D induced by Z — this variation is **"as good as" randomly assigned**.*

When Does IV Work?

IV solves endogeneity when:

- **Omitted variable** affects D but not Z
- **Measurement error** is in D, not Z
- **Simultaneity**: Z shifts D exogenously

IV does NOT solve when:

- Z is also correlated with omitted variables
- Z affects Y directly (**violates exclusion**)
- **Weak first stage** (Z barely affects D)

> *"**Key**: The **exclusion restriction** is **untestable** — it must be defended by theory!*

What is LATE?

LATE = **Local Average Treatment Effect**

$$\text{LATE} = E[Y_i(1) - Y_i(0) | \text{Complier}]$$

- The causal effect for **compliers only** — units whose treatment status changes because of the instrument
- **"Local"** = specific to the subpopulation affected by the instrument
- Different from **ATE** (Average Treatment Effect for entire population)

Setup: Binary instrument $Z \in \{0, 1\}$, binary treatment $D \in \{0, 1\}$; **four population types**:

TYPE	$D(0)$	$D(1)$	DESCRIPTION
Always-takers	1	1	Take treatment regardless
Never-takers	0	0	Never take treatment
Compliers	**0**	**1**	**Follow instrument**
Defiers	1	0	Ruled out by monotonicity

Understanding the Wald Formula

Wald estimator (Imbens & Angrist 1994):

$$\beta_{IV} = \frac{E[Y|Z=1] - E[Y|Z=0]}{E[D|Z=1] - E[D|Z=0]} = E[Y(1) - Y(0)|\text{Complier}]$$

Decomposition:

Numerator (Reduced Form): $E[Y|Z=1] - E[Y|Z=0]$

- Effect of scholarship offer on earnings
- **Zero** for always-takers and never-takers

Denominator (First Stage): $E[D|Z=1] - E[D|Z=0]$

- Effect of scholarship offer on college attendance
- Equals **share of compliers** in population

Ratio:

$$\frac{p_c \times \text{LATE}}{p_c} = \text{LATE}$$

> *IV gives the effect for the **"marginal"** population affected by the policy — often the **policy-relevant parameter**!*

Limitations of Classical IV

1. Homogeneous effects assumption

- Classical IV assumes θ is same for everyone
- Reality: Returns to education vary by ability, family background
- With heterogeneous effects, IV still works (**LATE**) but interpretation changes

2. Low-dimensional controls

- Must control for X linearly
- What if confounding is **non-linear**?
- What if we have **many controls** $(p > n)$?

3. Single effect

- Only gives **one number**, not **who benefits most**

PROBLEM	WHAT IS "HIGH-DIM"?	CORE ISSUE
High-dim controls	Many confounders X	Cannot partial out flexibly
High-dim IVs	Many instruments Z	Many-weak-IV bias, overfitting

> ***Next**: How ML addresses these limitations*

Part 3

High-Dimensional Challenge

Problem 1: High-Dimensional Controls

Modern empirical settings:

- Many potential confounders (demographics × geography × time)
- Text descriptions, images, embeddings
- Non-linear relationships

The problem:

- $p \approx n$ or $p > n$: Cannot include all controls in 2SLS
- Even if $p < n$, misspecified functional form biases estimates

Examples:

- **Returns to education**: Family background, ability, location, cohort → thousands of variables
- **401(k) wealth**: Income is highly non-linear; demographics create many interactions
- **AI demand**: Product images + text → 1,536-dimensional embeddings as controls

Naive approach fails: Lasso selects variables **predictive of** Y, missing confounders tied to D or Z.

> ***Belloni et al. (2012)***: *Must select from* ***ALL equations*** *(Y, D, and Z) — union of selected sets*

Post-Double Selection Lasso (Review)

Solution: Select controls relevant for **any part of the system**

Algorithm:

1. Lasso: Y on X → select S_1
2. Lasso: D on X → select S_2
3. Lasso: Z on X → select S_3
4. Union: $S = S_1 \cup S_2 \cup S_3$
5. Run 2SLS with controls in S

Intuition: Any variable that confounds Y, D, or Z is included

> *"**Limitation**: Still assumes linearity; **DML generalizes** to any ML method*

Problem 2: High-Dimensional IVs

Many instruments arise naturally in modern applications:

APPLICATION	STRUCTURE	NUMBER OF IVS
Angrist & Krueger (1991)	Quarter of birth × year × state	**180+** instruments
Judge designs	Judge fixed effects	**40-200+** instruments
Mendelian randomization	Genetic SNPs as IVs	**100-10,000+** instruments
Geographic IVs	Distance to many locations	**50-500+** instruments
Text-based IVs	Word counts / embeddings	**1,000+** instruments

The problem: With many instruments, 2SLS first stage overfits. Fitted values $\hat{D}_i$ become too close to actual D_i, transmitting endogeneity to the second stage.

> *"**Key reference**: Bekker (1994); Staiger & Stock (1994); Angrist, Imbens & Krueger (1999)*

Part 4

JIVE and Regularized IV

Jackknife IV (JIVE): The Overfitting Problem

Traditional 2SLS first stage:

$$D = Z\pi + v \quad \Rightarrow \quad \hat{\pi} = (Z'Z)^{-1}Z'D = \pi + \underbrace{(Z'Z)^{-1}Z'v}_{\text{correlated with every } v_i}$$

Fitted value for observation i (with leverage $h_{ii} = Z_i'(Z'Z)^{-1}Z_i$):

$$\hat{D}_i = Z_i'\hat{\pi} = Z_i'\pi + \underbrace{h_{ii}v_i}_{\text{leverage term}} + \sum_{j \neq i} h_{ij}v_j$$

The bias channel: $\hat{\pi} \leftarrow v_i \leftrightarrow u_i \Rightarrow \hat{D}_i$ correlated with second-stage error u_i

Jackknife fix: Estimate $\hat{\pi}^{(-i)}$ **without i**, then predict $\hat{D}_i^{(-i)} = Z_i'\hat{\pi}^{(-i)}$. This breaks the link.

JIVE Algorithm & UJIVE Correction

Step 1: Jackknife first stage — for each i, exclude obs i, fit $D_{-i} = \pi_0 + \pi_1 Z_{-i} + \epsilon_{-i}$, predict $\hat{D}_i^{(-i)}$

Step 2: Second stage — OLS of Y_i on $\hat{D}_i^{(-i)}$: consistent even with many instruments

Let $P_Z = Z(Z'Z)^{-1}Z'$ with diagonal leverage h_{ii}:

ESTIMATOR	FIRST-STAGE PREDICTION FORMULA	KEY PROPERTY
JIVE	$\tilde{D}_i = \frac{(P_Z D)_i - h_{ii} D_i}{1 - h_{ii}}$	Leave-one-out
UJIVE	$\tilde{D}_i = \frac{(P_Z D)_i}{1 - h_{ii}}$	Adjusts for leverage, better finite-sample

> *"**UJIVE** explicitly corrects for leverage via $\frac{1}{1-h_{ii}}$, eliminating small-sample bias from high-influence observations. Both solve overfitting via **sample splitting**.*

UJIVE vs RJIVE

Problem with classical JIVE/UJIVE: Still assumes **linear** first stage. When $p > n$, OLS overfits.

RJIVE (Regularized JIVE): Jackknife + regularized regression (Lasso/Ridge/Elastic Net) in first stage

	UJIVE	RJIVE
First stage	OLS + Jackknife	Regularized + Jackknife
$p > n$?	❌ No	✅ Yes
Sparse IVs	—	✅ Lasso
Dense IVs	—	✅ Ridge / Elastic Net
Use case	Many instruments, $p < n$	High-dimensional IVs

> *UJIVE corrects small-sample bias but stays low-dim. RJIVE scales to $p \geq n$.*

Sparse vs Dense IV Structure

When you have many potential instruments (p large), structure matters:

DIMENSION	SPARSE IV STRUCTURE	DENSE IV STRUCTURE
Signal pattern	Few strong IVs, many zeros	Many weak IVs, all non-zero
Example	3-4 strong instruments	100+ weak instruments
Intuition	"Needle in haystack"	"Many small streams"
β pattern	$\|\beta\|_0 \ll n$	$\|\beta\|_2$ bounded, most non-zero

> *This distinction determines which regularization to use!*

Sparse IV: Lasso-JIVE

Setting: True first stage is sparse — few strong instruments, many zero-coefficient noise instruments

Examples: Geographic IVs where only specific college distances matter; weather shocks with specific lags; genetic instruments (few valid SNPs)

Optimal approach: **Lasso (L1 penalty)**

- Selects the few strong instruments, ignores noise instruments

Dense IV: Ridge/PCA-IV

Setting: True first stage is dense — all $\beta_j \neq 0$ but each is small

Examples: Many weak instruments in finance; genetic risk scores; text-based instruments

Optimal approach: Ridge (L2 penalty) or **PCA**

PCA-IV steps:

1. Compute first k principal components from $Z_1, \ldots, Z_p$: $PC_j = \sum_{\ell=1}^{p} \phi_{j\ell} Z_\ell$
2. Use $PC_1, \ldots, PC_k$ as new instruments in 2SLS
3. Reduces dimensionality from p to $k \ll p$; avoids multicollinearity

> *"**Best for**: Many instruments that are **highly correlated** (genetic risk scores, text-based IVs, finance predictors)*

Choosing Your Method: Sparse vs Dense

STRUCTURE	METHOD	FIRST STAGE	WHEN TO USE
Sparse	Lasso-JIVE	$\|\hat{\beta}\|_1$ penalty	Few strong IVs, many irrelevant
Sparse	Post-Lasso JIVE	Select then OLS	Same, want valid inference
Dense	Ridge-JIVE	$\|\beta\|_2^2$ penalty	Many weak IVs, all relevant
Dense	PCA-IV	First k principal components	Many correlated instruments
Unknown	DML-IV	Cross-fit + any ML	Complex structure, non-linear

> ***How to decide?***: *Compare first-stage R^2 with* ***Lasso vs Ridge*** *on cross-validation.*

Empirical Examples: High-Dimensional IVs

STUDY	IV	# IVS	WHY VALID	KEY ISSUE
Angrist & Krueger (1991) *QJE*	Quarter of birth × year × state	180	QOB randomly assigned; compulsory laws link QOB to dropout age	Many weak IVs, jointly identify LATE
Judge FE (Mueller-Smith 2015)	Judge × offense × demo.	1,315	Quasi-random case assignment	Both high-dim IVs **and** 100+ controls
Mendelian randomization (Bockerman et al. 2019)	Individual SNPs for BMI	32–97	Randomly assigned at conception	Many-weak-IV; GRS collapses to avoid
Bartik / China Syndrome (ADH 2013) *AER*	Industry-share shift-share	400+ (equiv.)	Non-US industry shocks exogenous to US labor markets	Scalar Bartik = 400+ share instrs.
Cultural Revolution IVs (Fleisher & Wang 2005)	Cohort × region × CR-year	20–30	CR disruption by decree, not indiv. choice	Interaction terms individually weak

Empirical Examples: High-Dimensional Controls

STUDY	IV	CONTROLS	WHY HIGH-DIM	ML FIX
AJR (2001) *AER*	Settler mortality	Lat., coast, temp., continent FE + polys. (~50–100)	$p \approx n$ on 64 countries	DML-IV w/ RF / Lasso
401(k) wealth (Chernozhukov et al. 2018) *Econometrica*	401(k) eligibility	Income + demo. + non-linear terms (20+)	Income-wealth highly non-linear	DML-IV w/ neural net, boosting
AI demand (Chernozhukov et al. 2024)	Cost shifters	CLIP + BERT embs. ($p = 1{,}536$)	Unstructured product images + text	DML-IV + GRF
Digital infra. & income gap (Zhang & Zhao 2025)	Telephone penetration; terrain	City + year FE (~300 dummies)	$p \approx n$ on 3,135 obs	DML / PDS-Lasso

Part 5

DML-IV for High-Dimensional Controls

The DML-IV Setup

Model: Partially linear IV with high-dimensional controls

$$Y = \theta_0 D + g_0(X) + \zeta$$

$$D = r_0(X) + V$$

$$Z = m_0(X) + U$$

Where:

- θ_0: **Causal effect of interest** (target parameter)
- $g_0(X), r_0(X), m_0(X)$: **Unknown nuisance functions**
- X: High-dimensional controls (can be $p > n$)

Challenge: Estimate θ_0 with **valid inference** despite not knowing nuisance functions

The Core Idea: Residualization

Step 1: Partial out X using ML

- Predict Y from X: $\hat{g}(X) \approx E[Y|X]$, residual $\tilde{Y} = Y - \hat{g}(X)$
- Predict D from X: $\hat{r}(X) \approx E[D|X]$, residual $\tilde{D} = D - \hat{r}(X)$
- Predict Z from X: $\hat{m}(X) \approx E[Z|X]$, residual $\tilde{Z} = Z - \hat{m}(X)$

Step 2: Run IV on residuals

$$\hat{\theta} = \frac{\sum_i \tilde{Z}_i \tilde{Y}_i}{\sum_i \tilde{Z}_i \tilde{D}_i}$$

> ***Intuition***: *Residuals are **"purged"** of confounding from* X

Why Naive ML Fails (Regularization Bias)

Problem: If we estimate $\hat{g}, \hat{r}, \hat{m}$ on **same data**

- ML overfits (captures sample-specific noise)
- Residuals correlated with errors
- IV estimate **biased and inconsistent**

Error decomposition:

$$\sqrt{n}(\hat{\theta} - \theta_0) = \underbrace{\text{Normal}}_{\text{good}} + \underbrace{\sqrt{n} \times \text{Regularization Bias}}_{\text{bad}}$$

With ML converging at $n^{-1/4}$: bias $= \sqrt{n} \times n^{-1/4} = n^{1/4} \to \infty$

> ***Need solution**: Make estimator **insensitive** to nuisance estimation errors*

Neyman Orthogonality

Solution: Use a **score function** that is locally insensitive to nuisance errors

Definition: Score ψ is Neyman orthogonal if

$$\frac{\partial}{\partial \eta} E[\psi(W; \theta_0, \eta)]\Big|_{\eta=\eta_0} = 0$$

IV orthogonal score:

$$\psi = (Y - \theta D - g(X))(Z - m(X))$$

Why this works:

- Errors in $\hat{g}, \hat{m}$ have only **second-order** effect on $\hat{\theta}$
- Bias = $(\hat{g} - g)^2 \approx n^{-1/2}$ — negligible!
- $\sqrt{n}$**-consistency** restored

Cross-Fitting

Remaining problem: Even with orthogonal scores, some overfitting remains

Cross-fitting solution:

1. Randomly split data into K folds (typically $K = 5$)
2. For each fold k:
    - Train ML models on other $K - 1$ **folds**
    - Predict on fold k (**out-of-sample**)
3. Pool all out-of-sample residuals
4. Run IV on pooled residuals

> ***Key***: *Predictions are* ***out-of-sample*** *→ no overfitting correlation*

Connection to Jackknife IV: Both are **sample splitting** to break overfitting. Jackknife IV → UJIVE → DML generalizes to ML nuisances with fold-wise (vs. observation-wise) splitting.

DML-IV Algorithm Summary

Input: (Y, D, Z, X) with high-dimensional X

Step 1: Cross-fitting

- For each fold k, train ML on other folds
- Get out-of-sample predictions $\hat{g}_k, \hat{r}_k, \hat{m}_k$
- Compute residuals: $\tilde{Y} = Y - \hat{g}$, $\tilde{D} = D - \hat{r}$, $\tilde{Z} = Z - \hat{m}$

Step 2: Estimation

- Run IV on residuals: $\hat{\theta} = \frac{\sum \tilde{Z}_i \tilde{Y}_i}{\sum \tilde{Z}_i \tilde{D}_i}$
- Compute standard errors via bootstrap

> ***Output**: $\sqrt{n}$**-consistent** $\hat{\theta}$ with **valid inference***

Chernozhukov et al. (2018): Unifying Both Problems

How does DML-IV handle the two distinct high-dimensional problems?

1. High-dimensional controls (X is large)

- Use **any ML method** (RF, Lasso, NN) to estimate $g_0(X), r_0(X), m_0(X)$
- **Cross-fitting** ensures regularization bias does not contaminate the IV estimate
- **Neyman orthogonality** makes the estimator robust to nuisance estimation error

2. High-dimensional IVs (Z is large)

- DML-IV allows Z to be a vector; residualized first stage can itself be high-dimensional
- Combine with **regularization** in first stage (Lasso-JIVE, Ridge-JIVE) within cross-fitting
- The orthogonal score remains valid regardless of the dimensionality of Z

> ***Key insight***: *DML-IV separates* ***estimation*** *(flexible ML for nuisances) from* ***inference*** *(orthogonal score + cross-fitting).*

Weak Instruments in DML

Challenge: Flexible controls can **weaken instruments**

- Partialling out X aggressively removes variation
- Residual first stage: $\tilde{D}$ on $\tilde{Z}$ may have little power

Diagnostic: Check **residual first-stage F-statistic**

1. After cross-fitting, regress $\tilde{D}$ on $\tilde{Z}$
2. Rule of thumb: $F < 10$ indicates weak instruments

Solution: **Anderson-Rubin** confidence intervals

- Robust to weak instruments — **does not rely on first-stage estimates**
- **Intuition**: Under $H_0 : \theta = \theta_0$, $Y - \theta_0 D$ should be uncorrelated with Z. Test this directly.
- Invert the test statistic to find all θ_0 not rejected → valid CI
- Wider than standard CI but **valid regardless** of first-stage strength

> ***Why it works***: *Traditional t-statistic assumes strong first stage ($F > 10$). When F is small, the t-statistic is not normal. AR avoids this by testing the reduced-form relationship directly.*

Part 6

GRF-IV for Heterogeneous Effects

The Heterogeneous Effects Problem

Reality: Treatment effects vary across individuals

$$\theta_i = Y_i(1) - Y_i(0) \neq \text{constant}$$

Questions we want to answer:

- Who benefits most from education? (by ability, family income)
- Which firms respond most to subsidies? (by size, industry)
- How does price sensitivity vary by demographics?

Challenge: Estimate $\theta(x) = E[Y(1) - Y(0)|X = x]$ with valid inference

Generalized Random Forests for IV

Recall GRF (Lecture 6): Estimates parameters defined by **local moment conditions**

For IV, the moment condition at $X = x$:

$$E[(Y - \theta(x)D)(Z - E[Z|X = x])|X = x] = 0$$

GRF-IV algorithm:

1. Grow forest to obtain similarity weights $\alpha_i(x)$
2. Solve weighted IV equation in local neighborhood:

$$\hat{\theta}(x) = \frac{\sum_i \alpha_i(x)(Y_i - \bar{Y})(Z_i - \bar{Z})}{\sum_i \alpha_i(x)(D_i - \bar{D})(Z_i - \bar{Z})}$$

Weights $\alpha_i(x)$: From random forest — units with similar X get higher weight

> ***Result***: *Adaptive, data-driven heterogeneity; local Wald formula for each covariate point x*

GRF-IV vs DML-IV

FEATURE	GRF-IV	DML-IV
Target	Heterogeneous effects $\theta(x)$	Average effect θ
Output	Function $\hat{\theta}(x)$ for all x	Single number $\hat{\theta}$
Controls	Forest handles X automatically	Explicit nuisance estimation
Best for	Understanding **who benefits**	High-dim X, **average effect**
Package	`grf` (R)	`DoubleML` (Python/R)

> ***Connection***: *Both use ML to handle complex X, but for* ***different targets***

GRF-IV Example: Returns to Schooling

Setup: Card (1995) — distance to college as instrument for education

- Z = Distance to nearest college
- D = Years of schooling
- Y = Log wages
- X = Family income, ability proxies, location

Findings:

- $\theta(x)$ **higher** for low-income students (college access is binding constraint)
- $\theta(x)$ **lower** for high-ability students (would succeed regardless)
- $\theta(x)$ varies by **urban/rural** location

> *"**Policy insight**: Subsidies most **cost-effective** for disadvantaged populations*

Paper Deep Dives

AJR (2001) | Poterba, Venti & Wise | Chernozhukov et al. (2024)

Application 1: Acemoglu, Johnson & Robinson (2001)

"The The Colonial Origins of Comparative Development" *American Economic Review*, 2001

Research question: Do institutions (property rights, rule of law) cause economic growth?

Setup:

- Y = Log GDP per capita; D = expropriation risk; Z = settler mortality 1500–1800
- X = Geography (latitude, coast, temperature, continent FE) — polynomials & interactions ($p \approx 50–100$)
- Sample: only **64 countries** → $p \approx n$, linear controls insufficient

Original estimates (AJR 2001):

METHOD	$\hat{\theta}$	SE	INTERPRETATION
OLS (naive)	0.52	0.06	Biased (omitted variables)
2SLS (linear X)	0.94	0.16	Institutions → prosperity

Why DML-IV matters here: With $p \approx n$, flexible geography controls are essential but OLS/2SLS with polynomials suffers from overfitting. DML-IV (RF/Lasso) allows valid inference with high-dimensional, non-linear controls — estimates remain in the 0.8–1.0 range, validating AJR's main conclusion (Chernozhukov et al. 2018, §4.2).

Application 2: 401(k) and Wealth

"Do 401(k) Contributions Crowd Out Other Personal Saving?" Poterba, Venti & Wise (1995); reanalyzed in Chernozhukov et al. (2018)

Two estimands:

- **ATE of eligibility**: Effect of being *eligible* for 401(k) on net wealth (PLR)
- **LATE of participation**: Effect of actual *participation* on wealth, instrumented by eligibility (IIVM)

ATE results (eligibility):

METHOD	ESTIMATED EFFECT	NOTES
Naive OLS	$19,559	Biased by selection
DML (Lasso)	$8,975	Flexible income-wealth
DML (RF)	$8,910	Non-linear demographics
DML (XGB)	$8,597	Boosting, early stopping

Source: Reproduced from Chernozhukov et al. (2018, Econometrica, Table 3) via DoubleML documentation. 1991 SIPP; $ amounts in 1991 dollars.

Application 2: 401(k) and Wealth (continued)

LATE results (participation, IV with eligibility):

METHOD	ESTIMATED EFFECT	NOTES
DML-IV (Lasso)	$12,956	Compliers: no plan, no save
DML-IV (RF)	$12,003	Flexible first stage
DML-IV (XGB)	$13,258	Non-linear propensity

Source: Reproduced from Chernozhukov et al. (2018, Econometrica, Table 3) via DoubleML documentation. IIVM estimator; eligibility instrumented for participation.

> *DML cuts naive estimate by **~55%**; participation effect ($12k–13k) > eligibility effect ($8.6k–9k) because compliers are moderate-income households targeted by policy.*

Application 3: Bach et al. (2025)

"Adventures in Demand Analysis Using AI" — arXiv:2501.00382, 2025

Research question: What is the price elasticity of demand on Amazon, controlling for product quality?

Classic problem: Price $\leftarrow$ Quality $\rightarrow$ Demand — high-quality products command higher prices but sell more → naive elasticity **biased toward zero**

Methodology:

- Product images → BEiT (ViT) + text → RoBERTa/LLaMA → multimodal embeddings
- Cost shifters as instruments Z; **DML-IV** with embeddings as controls
- **GRF** to get heterogeneous elasticity $\epsilon(x)$ by product cluster

AI Demand: Results and Heterogeneity

METHOD [FEATURES]	Q_{it} R^2	P_{it} R^2
Linear Reg [tabular]	20.8%	15.8%
Boosted Trees [tabular]	47.5%	18.5%
Deep Learning [image + text + tabular]	**59.5%**	**66.7%**

Source: Bach, C., Hortaçsu, A., & Wildenbeest, M. (2025, arXiv:2501.00382, Table 2). Prediction R^2 for quantity (Q_{it}) and price (P_{it}) on Amazon toy-car listings.

Price elasticity estimates (toy cars, Amazon):

SPECIFICATION	$\hat{\epsilon}$	INTERPRETATION
Naive (no quality control)	≈ −0.1	Implausibly inelastic
With lagged rank $Q_{i,t-1}$	≈ −0.7 (rank) → **−1.4** (demand)	Realistic range
DML with AI embeddings	−4.0 to 0.8	Strong heterogeneity by product cluster

Source: Bach et al. (2025, arXiv:2501.00382, Table 4 & Figure 5). Elasticity estimates via DML with product-embedding controls; heterogeneity across product clusters.

> *AI embeddings capture quality/branding; omitting them **conflates** price effect with quality effect. DML enables use of **unstructured data** for causal inference.*

Summary: Choosing Your Method

SITUATION	RECOMMENDED METHOD	PACKAGE
Low-dim X, homogeneous effects	**2SLS**	`linearmodels`
Many instruments, sparse structure	**Lasso-JIVE**	`hdm`
Many instruments, dense structure	**Ridge-JIVE / PCA-IV**	`sklearn`
High-dim X, want average effect	**DML-IV**	`DoubleML`
Want heterogeneous effects $\theta(x)$	**GRF-IV**	`grf`
Weak instruments suspected	**DML-IV + AR tests**	`DoubleML`
Unstructured data (text/images)	**DML-IV with embeddings**	`DoubleML` + `transformers`

> ***Key insight**: ML doesn't replace IV logic — it enhances our ability to control for complex confounders and estimate heterogeneous effects*

Key Takeaways

1. IV identifies LATE

> *Effect for **compliers**, not population average*

2. Exclusion is key

> *Must defend $Z \to Y$ only through D via **theory***

3. High-dim IVs

> *JIVE/RJIVE handle many instruments via **jackknife + regularization**; choose **Lasso for sparse**, **Ridge for dense***

4. High-dim controls

> ***DML** fixes classical IV via **orthogonality + cross-fitting***

5. Heterogeneity

Further Reading

PAPER	AUTHORS	KEY CONTRIBUTION
Identification & Estimation of Causal Effects	Imbens & Angrist (1994), *Econometrica*	**LATE framework**
Jackknife Instrumental Variables Estimation	Angrist, Imbens & Krueger (1999), *JASA*	**JIVE method**
Double Machine Learning for IV	Chernozhukov et al. (2018), *Econometrica*	**DML-IV theory**

Further Reading (continued)

PAPER	AUTHORS	KEY CONTRIBUTION
Generalized Random Forests	Athey, Tibshirani & Wager (2019), *Annals*	**GRF for het. effects**
The Colonial Origins of Comparative Development	Acemoglu, Johnson & Robinson (2001), *AER*	**Classic IV application**
Adventures in Demand Estimation	Chernozhukov et al. (2024)	**AI embeddings for demand**

Data and Code

This lecture:

- Slides source: `_slides/Lecture08-Instrumental Variables and DML-IV-Slides.md`
- Exercises: `exercises/Lecture08-Exercise.ipynb`

Key packages:

- Python: `econml` (DML-IV), `ivmodels`

Datasets:

- Angrist & Evans (1998) sibling sex composition; 401(k) IV

Thank You!

Next Lecture: Difference-in-Differences and RDD